(K.A.N.I.N)—Knowledge-driven Agriculture for Nurturing and Intelligent Nutrition System To Track Harvests and Plantations of Rice (*Oryza Sativa)*

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To NCF

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INTRODUCTION

Rice (*Oryza sativa)* is one of the staple foods in the world, feeding more than half of the global population, especially in Asia, where it is a major part of the daily diet in many countries. In the Philippines, agriculture plays a significant role in the country's economy, providing a livelihood for farmers and ensuring food security for the nation. However, the sector continues to struggle due to labor shortages, unpredictable weather, and inefficient farming practices that reduce productivity. As climate change intensifies, Filipino farmers face increasing challenges such as droughts, floods, and soil degradation, which threaten both yields and income stability. Recent advancements in robotics and artificial intelligence (AI) have opened

opportunities to modernize agriculture through automation and data-driven decision-making.

While Internet of Things (IoT)-based systems have already been used to monitor crops and soil conditions, most rely on static thresholds and lack adaptive intelligence. This gap inspired the development of an AI-enhanced crop growth monitoring machine called the Knowledge-driven Agriculture for Nurturing and Intelligent Nutrition system, or simply, K.A.N.I.N.—a web-based automation platform that combines environmental sensing, real-time data visualization, and predictive machine learning.

This study aims to develop a web-based automation device utilizing modern farming technologies to address the financial challenges faced by Filipino farmers and modernize traditional farming practices. By integrating sensors and data analytics. The system seeks to promote a much more precise agriculture that enables farmers to optimize their crop cultivation processes, enhance productivity and yield, and ultimately improve their economic sustainability and quality of life.

Agriculture remains a fundamental pillar of the Philippine economy, providing livelihood to millions of farmers and ensuring national food security. Despite its importance, the sector continues to face numerous challenges, including labor shortages, unpredictable weather patterns, and inefficient farming practices that

hinder productivity. The growing impacts of climate change, manifested through prolonged droughts, soil degradation, and temperature fluctuations, further exacerbate these challenges, posing significant threats to crop yield stability and the overall resilience of the agricultural sector.

Statement of the Problem/Objectives

Traditional farming in the Philippines often relies on manual and experience-based decision-making, which can result in inefficient resource use and reduced crop yield. Although several IoT-based monitoring systems have been developed to assist in crop management, most existing models are limited to basic data collection and lack intelligent analysis and real-time responsiveness. This study seeks to design and develop the Knowledge-driven Agriculture for Nurturing and Intelligent Nutrition (K.A.N.I.N.), a web-based crop growth monitoring system that utilizes IoT sensors to gather and analyze environmental data essential for rice cultivation. This research aims to address inefficiencies in traditional monitoring systems by providing a platform for data representation.

Engineering Goals

To design and implement a robust and efficient Knowledge-driven Agriculture for Nurturing and Intelligent Nutrition System (K.A.N.I.N System) for rice cultivation, and to evaluate its performance in terms of data accuracy, system reliability, real-time monitoring capabilities, and impact on rice yield optimization during cultivation and harvesting. This system will integrate environmental sensor data to enhance decision-making processes and improve overall agricultural productivity. All while being eco-friendly with the use of solar energy. The device must cost less than ₱20,000 pesos and be portable. It aims to integrate IoT-based sensors, including soil moisture, temperature, soil nutrient level, humidity, pH, and light intensity sensors into a durable and weather-resistant structure capable of continuous field monitoring.

Significance of the Study

The development of the knowledge-driven Agriculture for Nurturing and Intelligent Nutrition (K.A.N.I.N.) holds significant importance in advancing sustainable and technology-driven rice farming in the Philippines. By integrating web-based automation, real-time environmental sensing, and intelligent data analysis, this research aims to address long-standing challenges in the agricultural sector.

For farmers, the system provides a practical and accessible tool for monitoring essential environmental factors such as soil moisture, pH, temperature, and humidity sensors. Through continuous data collection and analysis, K.A.N.I.N. enables

informed decision-making for crop health management, ultimately improving yield quality and reducing input costs.

For researchers and agricultural technologists, this study contributes to the growing body of knowledge on IoT-based and web-enabled precision agriculture systems. The system's modular design allows for further enhancement and adaptation across different crops and farming conditions, promoting innovation and scalability within the agricultural technology field.

For policymakers and government agencies, K.A.N.I.N. demonstrates the potential of integrating smart farming technologies into national agricultural development programs that align with the country's goals.

Scope and Delimitations

The general intent of this research is to design, develop, and evaluate the Knowledge-driven Agriculture for Nurturing and Intelligent Nutrition (K.A.N.I.N.), a web-based crop growth monitoring system that utilizes Internet of Things (IoT) technology to enhance rice (Oryza sativa) cultivation. The study focuses on the integration of various sensors—including soil moisture, soil temperature, pH, humidity, light intensity, and nutrient sensors—into a centralized web platform that allows farmers to monitor environmental parameters in real time.

The study will involve the deployment of IoT-based sensors connected to an ESP32 microcontroller, designed to collect environmental data from a controlled rice

field setup. The collected data will be transmitted to a web-based monitoring interface, where information is processed, displayed, and stored for future analysis. The main objectives include testing the system's performance in terms of data accuracy, reliability, responsiveness, and operational efficiency.

This research will be conducted in Naga City, Camarines Sur, where the monitoring prototype will be tested under actual environmental conditions. The system's performance will be evaluated through controlled observations and recorded data accuracy comparisons.

The study will exclusively focus on the environmental monitoring aspect of rice farming, particularly parameters that influence crop growth, such as soil condition, temperature, humidity, and light exposure. It will not cover pest management, weed detection, or large-scale farm automation. The testing phase will be limited to small- and medium-scale rice plots, and will not involve the full automation of irrigation or fertilization systems. Moreover, the predictive functions of K.A.N.I.N. will rely solely on sensor-based data inputs and will not include satellite imaging or drone-based data collection.

Finally, factors beyond the control of the researchers—such as extreme weather conditions, unstable internet connectivity, and power interruptions—are

excluded from the scope of this study and may affect the accuracy of data transmission or system performance.

METHODOLOGY

Research Design

The study aims to test the Knowledge-driven Agriculture for Nurturing and Intelligent Nutrition System (K.A.N.I.N. System), designed to automate rice field monitoring and optimize environmental conditions for crop growth. The system will utilize a combination of sensors, Arduino-based hardware, and IoT integration to collect, transmit, and analyze real-time data such as soil moisture, pH, temperature, soil nutrient level, humidity, and light intensity. Through automated responses and visual data dashboards, the K.A.N.I.N. System aims to enhance precision agriculture by reducing manual labor, improving resource efficiency, and increasing crop yield.

Materials/Equipment

The materials we will be using include ESP32 DevKit, Soil temperature sensor, Soil moisture sensor, Air humidity sensor & Temp sensor, pH sensor, and DF robot NPK sensor. It continuously collects and displays environmental data to inform farmers and help them make data-driven decisions for better crop growth, yield, and management.

System Design:

The system is programmed to monitor factors needed for rice (*oryza sativa)* to grow in optimal conditions and intends to inform the user of any imbalance in the environmental factors that play major/minor roles in the crop's growth.

- **DS18B20 Soil Temperature Sensor**:

Monitors soil temperature to support optimal conditions for rice growth. We will only be using 1 of these sensors. We chose this sensor because it is cost-efficient, accurate, and waterproof, and not only these, but also because it has the appropriate pins to connect to the ESP32.

- **Capacitive Soil Moisture Sensor V2.0**

Measures soil moisture to ensure proper irrigation and prevent water wastage. We had a wide range of soil sensors we could pick, but this caught our attention the most since it is capacitive; it is able to store electric discharge. The sensor is coated with a plate that acts like one side of a capacitor, thus making it more stable and durable.

- **Gravity: Analog pH Sensor Kit**

Assess soil pH to maintain optimal acidity for nutrient availability and plant growth. We personally chose to use this sensor because it is more accurate than some sensors out there, and the readings of this sensor are consistent; because of this, this makes the sensor a reliable tool for measuring the pH level of the soil.

- **SKU: SENO605 NPK Sensor**

Measures the level of nitrogen, phosphorus, and potassium. To assess the crop's fertility for the crop’s optimal growth. We chose this sensor because of its high-accuracy readings, fast response, waterproofing, corrosion resistance, and durability. These PROs made this sensor our top pick.

- **DHT11 Air humidity and temperature sensor**

Tracks air temperature and humidity to assess environmental conditions for plant health. We picked this sensor because of its compact structure; this sensor may look small, but it measures 2 things from the air: the humidity and temperature. Due to its size, it made it easier for us to create a design in which it would fit. Not only is its performance very good, considering its low price.

- **ESP32**

A Wi-Fi- and Bluetooth-enabled microcontroller is used to wirelessly connect and transmit data from the monitoring system. We picked this as our microcontroller because of its unique dual-core processor, integrated Wi-Fi and Bluetooth, and large number of peripherals; because of these, it is suitable for IoT projects that we will implement. Because of these integrations, it can do complex tasks while staying connected to the net with any other hardware pieces.

- **12 V 3000 mAh Ni-MH Pack for YionFestool/Work Tools**

A rechargeable battery pack rated at 12 volts and 3000 mAh (3.0 A.h), which means in ideal conditions, it stores about 36 Wh (12 x 3.0) of energy. It uses NiMH (Nickel-Metal Hydride) chemistry. It is built for tool use, which is perfect for our monitoring system, and can operate for 9-12 hours of straight usage. We chose not to use lithium-based batteries since we will be dealing with agricultural plots that involve water, and lithium reacts with water, causing an explosion. Also NiMH is rechargeable, reusable, environmentally safe, high capacity, has good current output, has a lower risk of explosion, fire, or any reaction, and is cost-effective.

- **50w 12v polycrystalline solar panel**

This collects sunlight and converts it into electrical energy by absorbing photons. We chose this solar panel, specifically polycrystalline, since it was more affordable and efficient for its price, and it also has the right measurements, making it fit for the spot for the solar panel on top of the prototype.

Procedures

1. **Hardware Setup**

Each sensor is connected to the ESP32 according to its pins (RX/TX). Sensor calibration will be conducted to ensure data accuracy and reliability. It will be powered by a rechargeable Nickel-Metal Hydride (NiMH) and a 0.020 kWh solar panel 0.020kWh.

2. **Programming/Coding**

Firmware will be built for the ESP32 microcontroller, sensors, and other components for data collection, transmission, and analysis. We will also integrate the ESP32 Wifi module to send real-time sensor data to BlynkIoT (cloud-based storage).

3. **Sensor Calibration and Testing**

All of the sensors will be calibrated to ensure accurate data reading. Data shall be collected within an interval of 15 minutes.

pH sensor: calibrated using buffer solutions (pH levels 4, 7, 10)

Soil moisture sensor: calibrated using 3 samples of soil (1 dry sample, 1 wet sample, and 1 moist sample).

Soil temperature sensor: calibrated using a digital or laboratory-grade thermometer as a reference to ensure they both have the same temperature reading.

Air humidity and temperature sensor: calibrated using a hygrometer and thermometer (lab grade) as a reference.

Data Gathering

The data-gathering procedure will focus on evaluating the performance, accuracy, and functionality of the K.A.N.I.N. System through continuous monitoring and analysis of environmental and operational parameters. Data collection will be performed using the stationary robotic prototype equipped with multiple sensors integrated into the ESP32 microcontroller system.

The system will collect environmental data, including soil moisture, soil temperature, air temperature, humidity, light intensity, and soil pH. These sensors will transmit real-time readings to the ESP32, which will process and log the data on an SD card while simultaneously sending it to a cloud-based IoT platform through the ESP32 module. The data will be updated at fifteen-minute intervals to ensure consistency and accuracy in recording environmental changes.

Each reading will be timestamped with the Real-Time Clock (RTC) module, allowing for correlation between data trends and specific timeframes of crop growth or environmental fluctuation. This will enable the system to track how temperature, moisture, or light intensity vary throughout the day and how these affect plant development.

The system performance data will also be monitored, including transmission rate, power consumption, response time, and data accuracy. These metrics will help assess the reliability of the K.A.N.I.N. system under different field conditions. Accuracy tests will be conducted by comparing sensor data with standard laboratory instruments or manual measurements to verify calibration and ensure data validity.

Additionally, operational data such as real-time data response and overall system stability will be recorded. The robot will be tested in both indoor controlled environments and outdoor rice plots to evaluate its adaptability and durability under varying weather conditions.

The collected data will then be processed, tabulated, and visualized through charts and graphs using the IoT dashboard and statistical software. Trends will be analyzed to identify relationships between environmental conditions and crop health indicators. This analysis will serve as the basis for determining the system's accuracy, efficiency, and potential contributions to improved rice cultivation practice

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APPENDIX A.

Definition of Terms

Automation—the utilization of technology to execute activities with minimum human involvement

Blynk IoT Platform—An accessible IoT platform enabling developers to prototype, deploy, and remotely manage interconnected devices, as well as to design bespoke mobile and web applications.

Calibration—the process of configuring and verifying the accuracy of a measuring instrument to ensure that its readings correspond to a recognized standard.

Cloud Storage—a service that allows users to store digital data on remote servers instead of a local hard drive and can be accessed from any device with an internet connection.

Crop yield—a measure of agricultural productivity, specifically the amount of crop harvested per unit of land area, expressed as kilos per hectare or bushels per acre.

Data analytics—the act of gathering, transforming, and organizing information to detect trends, answer questions, and make informed decisions.

Data visualization—the discipline of transforming data into visual contexts, such as charts, graphs, and maps, to help the human brain understand and discover insights.

DHT11—a low-cost digital sensor that monitors both ambient air temperature and relative humidity. It detects changes in the surrounding environment using a capacitive humidity sensor and a thermistor and then converts them into a calibrated digital signal that a microcontroller can interpret.

Economic sustainability—the practice of providing long-term economic benefit while safeguarding the environment and maintaining the well-being of future generations.

Environmental sensor—a device that detects, measures, and monitors physical, chemical, or biological characteristics in the environment, such as temperature, humidity, air quality, and light.

ESP32—a low-cost, low-power microcontroller series with built-in Wi-Fi and Bluetooth connection, aimed particularly for Internet of Things (IoT) and mobile applications.

Food Security—When all people have constant physical, social, and economic access to sufficient, safe, and nutritious food that fits their dietary needs and preferences for an active and healthy life.

Internet of Things (IoT)—a network of physical objects embedded with sensors, software, and other technologies that enable them to connect and exchange data with each other and the internet.

K.A.N.I.N. (Knowledge-driven Agriculture for Nurturing and Intelligent Nutrition)—a web-based automation platform that combines environmental sensing, real-time data visualization, and predictive machine learning. Please refer to our Introduction, Statement of the Problem, Scope and Limitations, and Methodology.

Machine Learning—the usage and development of computer systems capable of learning and adapting without explicit instructions, utilizing algorithms and statistical models to analyze and draw conclusions from data patterns.

NPK sensor—a device that monitors the levels of nitrogen (N), phosphorus (P), and potassium (K) in soil to evaluate its fertility. The sensor collects data from probes implanted into the soil, allowing farmers and gardeners to make informed decisions about applying fertilizer, optimizing crop yields, and limiting environmental effect. These sensors frequently use an electrical conductivity or optical approach and are designed to withstand long-term use in a variety of soil types.

pH sensor—An electronic device that detects the concentration of hydrogen ions in a liquid to determine its acidity or alkalinity.

Precision Agriculture—a data-driven, technology-enabled approach to farming that optimizes resource utilization while increasing agricultural yields and farm profitability.

Real-time monitoring—the continuous observation, measurement, and analysis of data or processes as they occur.

Resource efficiency—entails reducing the environmental effect of materials, water, and energy consumption across a product's life cycle, from extraction to disposal.

Smart farming—the use of advanced technology like the Internet of Things (IoT), drones, and artificial intelligence (AI) to manage farms more efficiently, boost food yields, and reduce resource use.

Soil Moisture Sensor—monitors soil conditions for agricultural irrigation, forestry, and environmental monitoring, assisting in optimizing watering schedules and preventing drought or flooding.

Soil pH—a measure of how acidic or alkaline soil is, with a range of 0 to 14.

Soil Temperature Sensor (DS18B20)—a digital temperature sensor that may be submerged in soil to properly monitor the temperature. It has a unique "one-wire" interface, requiring only one data line to connect with a microcontroller, and is frequently available in a waterproof probe form, making it excellent for usage in damp or severe situations, such as soil.

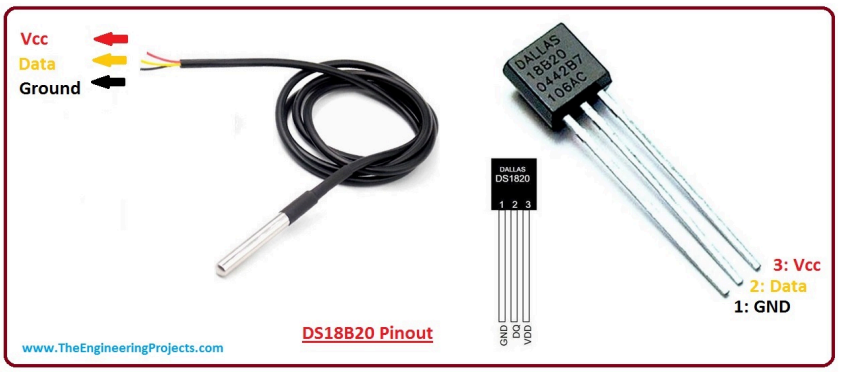
Solar energy—the power of the sun captured as heat or electricity, and it is a clean, renewable energy source.

Sustainable agriculture—an agricultural practice that fulfills present demands while preserving the ability of future generations to meet their own.

Web-based interface (GUI)—a visual component of an online application, allowing users to interact with the website through a web browser. It includes features like buttons, forms, menus, and navigation, ensuring usability and ease of use for users.

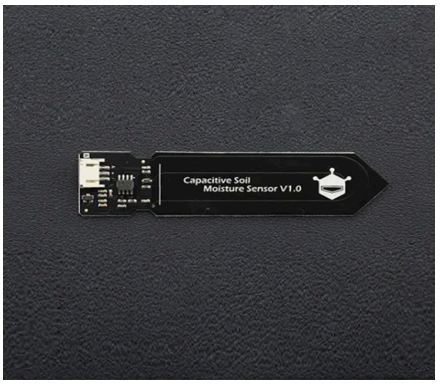
APPENDIX B - **EQUIPMENT USED**

Figure 1. DS18B20



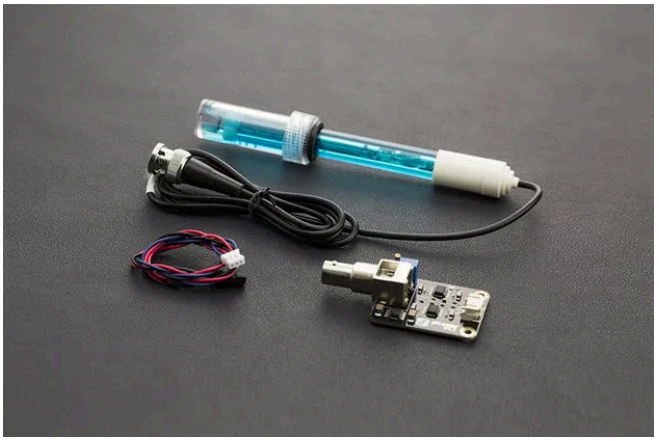
- A digital temperature sensor used to measure temperature accurately. The sensor provides reliable temperature readings and is commonly used in environmental monitoring applications. In this study the DS18B20 will be used to collect temperature data for system monitoring and analysis.

Figure 2. Capacitive Soil Moisture Sensor v2.0



- An electric sensor used to measure soil moisture content by detecting changes in soil dielectric permittivity and operates without exposed conductive probes to avoid corrosion and increase durability for long term use. In this study, the capacitive soil moisture sensor v2.0 will be used to monitor soil moisture conditions and support data-driven agricultural analysis.

Figure 3. Gravity: Analog pH sensor kit



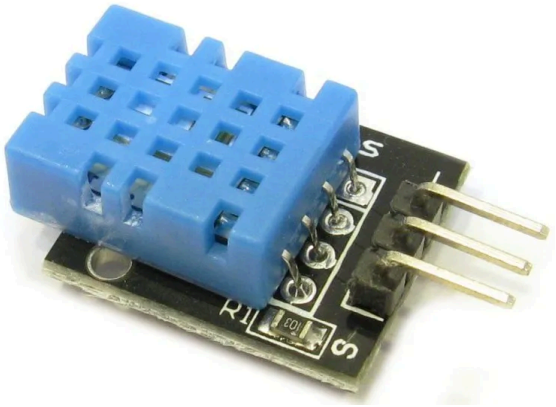
- An electronic sensor used to measure soil acidity or alkalinity by detecting hydrogen ion concentration through an analog signal. In this study, the Gravity analog pH sensor kit will be used to monitor soil pH levels and support data-driven soil conditions.

Figure 4. SKU: Seno605 NPK Sensor



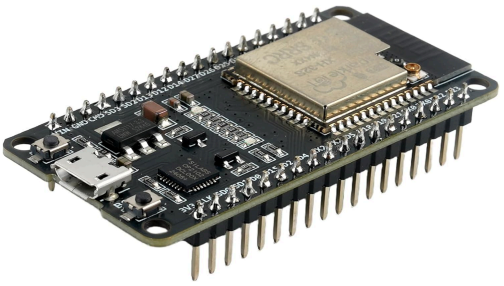
- An electronic soil analysis device designed to measure the concentration of nitrogen (N), phosphorus (P), and potassium (K) in soil. It analyzes soil electrical conductivity to estimate nutrient levels. In this study, the SKU: Seno605 NPK sensor will be used to monitor soil nutrient levels.

Figure 5.DHT11 Air humidity and temperature sensor



- A digital environmental sensor designed to measure ambient air temperature and relative humidity using a calibrated internal sensing element. In this study, the DHT11 sensor will be used to monitor atmospheric conditions affecting crop growth.

Figure 6.ESP32



- Is a low-power microcontroller with built in Wi-Fi. Also widely used in embedded systems and Internet of things (IoT) applications due to its high processing speed and multiple I/O (input/output)

Figure 7.12 V 3000 mAh Ni-MH Pack for Yion/Festool/Work Tools



- A rechargeable power storage device designed to supply stable electrical energy to electronic systems. In this study, the Ni-MH battery pack will be used to provide a safe and reliable power source for continuous field operation.

Figure 8.50w 12v Polycrystalline solar panel



- A photovoltaic device designed to convert sunlight into electrical energy using polycrystalline silicon cells. In this study, the solar panel will be used to

supply renewable energy and support sustainable operation of the monitoring system.

APPENDIX C - **COST ANALYSIS**

Item	**Quantity**	**Estimated Cost (₱)**
ESP32 Development Board	1	200
DS18B20 Soil Temperature Sensor	1	150
Capacitive Soil Moisture Sensor V2.0	1	400
DHT11 Air Temperature & Humidity Sensor	1	100
Analog pH Sensor Kit	1	600-1500
12 V 3000 mAh Ni-MH Battery Pack	1	800-1300
50 W 12 V Polycrystalline Solar Panel	1	2000
SD Card Module + SD Card (4–8 GB)	1	300
RTC Module (DS3231)	1	100
Total no. of materials and cost (Estimate)	9	6050